
Instructions. (15 points) Sections 14.8, 15.1, 15.1b. Show relevant work for full credit.

- (4^{pts}) **1.** Use Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 2x + y$ subject to the constraint $x^2 + y^2 = 5$.

- (4^{pts}) **2.** Approximate $\iint_R (x + 3y) dA$, where $R = [1, 3] \times [0, 6]$, by using a Riemann sum with a 2×2 grid of equal subrectangles and **midpoints** as sample points.

(3^{pts}) **3.** Without performing any integration, evaluate $\iint_R (3 - x) \, dA$ where $R = [0, 3] \times [0, 2]$.

Hint: Sketch the solid and identify its geometric shape. You may verify your answer using an iterated integral if you wish.

(4^{pts}) **4.** Evaluate $\int x \cos x \, dx$.